

Data Professional Survey Dashboard | Power BI

Project Overview

This project analyzes survey responses from 630 data professionals worldwide to understand current trends in the data industry. The dashboard provides insights into salary distributions, job roles, demographic characteristics, programming language preferences, and workplace satisfaction.

Using Power BI, an interactive dashboard was developed to transform raw survey data into actionable business insights and industry benchmarks.

Data Used

Dataset - Data Professional Survey Dataset

Number of Records - 630 Survey Responses

Data Visualization - Power BI

Business Questions

1. Which countries contribute the largest number of survey respondents?
2. How do average salaries vary across data-related job titles?
3. What is the demographic profile of survey participants?
4. Which programming languages are most preferred among data professionals?
5. How satisfied are professionals with their work-life balance?
6. How satisfied are professionals with their current salary?
7. Which data careers offer the highest compensation?
8. What trends can be identified within the global data workforce?

Dashboard Preview



Key Insights

Workforce Demographics

- A total of 630 data professionals participated in the survey.
- The average respondent age is 29.87 years.
- The largest respondent groups came from Canada, India, the United States, and the United Kingdom.

Salary Analysis

- Data Scientists reported the highest average salary among surveyed occupations.
- Data Engineers and Data Architects followed closely behind.
- Students and job seekers recorded the lowest average salary levels.

Programming Language Preferences

- Python is the dominant programming language across nearly all data-related professions.
- R remains the second most popular language, particularly among analytical and statistical professionals.
- Java, JavaScript, and C/C++ showed significantly lower adoption rates.

Workplace Satisfaction

- Average work-life balance satisfaction scored 5.74 out of 10.
- Average salary satisfaction scored 4.27 out of 10.
- Compensation satisfaction is substantially lower than work-life balance satisfaction, suggesting salary remains a major concern among data professionals.

Industry Trends

- Python continues to be the primary technical skill across the data industry.
- Advanced technical roles generally receive higher compensation.
- Despite strong demand for data talent, overall salary satisfaction remains relatively low.

Limitations

- Survey responses are self-reported and may contain personal bias.
- The dataset contains only 630 respondents and may not fully represent the global data professional population.
- Salary comparisons do not account for geographic cost-of-living differences.
- Some job categories have relatively small sample sizes, which may affect average salary estimates.
- Survey findings represent a specific point in time and may change as industry conditions evolve.